Multiplication law

1 Simplify:

a $y^3 \times y^2$ **b** $x^6 \times 8x^3$ **c** $3y^6 \times 4y$ **d** $x^y x^z$

e $a^7 \times -6a^5$ **f** $-2a^4 \times 5a$ **g** $-b^7 \times -b^6$ **h** $a^b a^c$

i $5y^5 \times 6y^3$ **j** $-8u^3 \times -3u$ **k** $-4b^6 \times -5b^7$ **l** $x^a x^b x^c$

2 Complete the following statements:

a $b^4 \times b^{\square} = b^7$ **b** $7x^{15} \times \square = 35x^{27}$

c $c^{10} \times c^{\square} = c^8$ **d** $4x^{20} \times \square = 24x^{25}$

Division law

3 Simplify:

a $a^9 \div a^5$ **b** $\dfrac{x^{11}}{5x^8}$ **c** $\dfrac{6m^{15}}{m^4}$ **d** $b^6 \div b^2$

e $\dfrac{j^9}{j^2}$ **f** $x^7 \div x^4$ **g** $\dfrac{f^8}{f^5}$ **h** $s^{13} \div s^4$

i $\dfrac{4k^{11}}{2k^6}$ **j** $\dfrac{8g^{12}}{g^5}$ **k** $6p^{13} \div 2p^{10}$ **l** $15l^8 \div 3l^2$

4 Complete the following statements:

a $b^9 \div b^{\square} = b^5$ **b** $x^{\square} \div x^3 = x^2$

c $\dfrac{g^{\square}}{g^4} = g^7$ **d** $\dfrac{k^7}{k^{\square}} = k^3$

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Power of a power law

5 Consider the expression $(p^3)^2$.

a State whether the following expressions are equivalent to $(p^3)^2$:

i $(p \times p \times p) \times (p \times p \times p)$

ii $p^3 \times p^3$

iii $(p \times p \times p)^2$

iv $p^3 \times p^2$

v $(p \times p \times p) \times (p \times p)$

b State whether the following equations are true or false:

i $(p^3)^2 = p^{3+2}$

ii $(p^3)^2 = p^{3\times 2}$

c Complete the following:

$$(p^3)^2 = p^{\square}$$

6 Simplify:

a $(j^3)^6$

b $(w^2)^4$

c $(t^4)^3$

d $(3a^4)^3$

e $(5h^7)^3$

f $(xy^2)^5$

g $\left(\frac{2}{h^2}\right)^4$

h $\left(\frac{a^2}{b^5}\right)^3$

i $\left(\frac{3x^2}{y^5}\right)^3$

j $(-x^9)^4$

k $(-5x^4)^3$

l $\left(\frac{-2a^4}{3b^2}\right)^3$

7 Find the value of a and b in the following equation:

$$\frac{v^{18}}{w^{24}} = \left(\frac{v^a}{w^4}\right)^b$$

Zero index

8 Simplify:

a $18a^0$

b $(f^0)^9$

c $(g^{12})^0$

d $(6a)^0$

e $8r^0 + (2q^3)^0$

f $\left(\frac{2}{x^2}\right)^0$

g $\frac{3r^0}{t^2}$

h $8k^0 \div (4x^2)^0$

9 Complete the following statements:

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a $b^{11} \div b^{11} = b^{\square}$ **b** $\dfrac{h^{\square}}{h^8} = h^0$ **c** $\dfrac{v^9}{v^{\square}} = v^0$ **d** $y^{\square} \times y^5 = y^5$

Mixed laws

10 Simplify:

a $\left(x^6 y^3\right)^4$

b $\dfrac{y^7 \times y^6}{y^3 \times y^2}$

c $p^5 \times \left(p^4\right)^3$

d $\left(2y^2\right)^3$

e $m^5 \div m^2 \times m^5$

f $p^9 \div p^5 \div p^2$

g $\left(-2x^3\right)^4$

h $\dfrac{6p^5 \times 4p^7}{8p^3}$

i $\left(7x^2\right)^0 - \left(8x^6\right)^0$

j $\left(12x^4\right)^0 + 12^0 - 12h^0$

k $\left(4h^0\right)^2 + 18 \div \left(3g^0\right)$

l $\left(\left(x^2\right)^6\right)^5$

m $\left(4u^5 v^2\right)^2$

n $\left(3a^2 b^5 c\right)^4$

o $\left(3y^5\right)^2 \times \left(5y^2\right)^3$

p $\dfrac{\left(x^4\right)^2}{x^5}$

q $\dfrac{\left(2x^2 y^0\right)^4}{x^5}$

r $\left(c^{10}\right)^{11} \div \left(c^8\right)^3$

s $\dfrac{24a^3}{(2a)(4b)}$

t $\left(10x^2 \times 10x^5\right)^0 - 10x^0$

u $\dfrac{3^{4a+2} \times 3^{1+6a}}{\left(3^3\right)^{3a-1}}$

v $\dfrac{81^{7a-4} \times 9^{3a+2}}{27^{3-3a}}$

11 Write $\left(16^p\right)^4$ in the form a^b, where a is a prime number.

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